Subliminal Prompting Beyond Static Geometry: Causal Depth and Multi-Token Confounds

Anonymous submission

Abstract

Subliminal learning shows that language models can transmit traits through apparently unrelated outputs. Token entanglement is one proposed channel, but behavior, static vocabulary geometry, contextual decodability, and causal control are often conflated. We separate them in a fixed animal-number prompting protocol. From Llama-3.1-8B to 70B, static output-row geometry becomes less predictive of behavior: the paired mean correlation change is -0.080 (95% CI [-0.127, -0.035]). A fixed output-head readout shows no resolved change in normalized depth AUC. Natural-state patching gives a different result. Conditional donor-control AUC rises from 0.254 to 0.540, a paired change of $+0.286$ (95% CI [$+0.272$, $+0.300$]), with increases for all 18 concepts. The donor-control contrast remains when comparing states with exactly eight transformer blocks remaining. Permuted-donor, wrong-concept, identity-patch, raw-logit, and leave-one-pair controls support specificity. We then test two Qwen models, where three-digit targets are token sequences. Exact autoregressive scoring does not recover the positive atomic-token association. Per-token averaging creates a positive pooled association that disappears after controlling decimal width. Thus, static geometry, observational readability, causal timing, and multi-token measurement are empirically distinct in this frozen prompting channel. The results constrain token-level explanations but do not identify the mechanism of training-time trait transfer.

1 Introduction

Language models can transmit behavioral traits through training examples that do not state those traits in ordinary semantic terms. In a canonical example, a teacher prompted to prefer owls emits lists of numbers; a student fine-tuned on those lists later exhibits an owl preference (Cloud et al. 2025). This phenomenon, called *subliminal learning*, motivates a precise measurement question: what signal connects an apparently unrelated output to the prompted concept?

Token entanglement proposes a coupling between vocabulary items, internal representations, and behavior (Zur et al. 2025). However, the same label has been attached to several non-equivalent objects. An animal and a number can be correlated in output behavior, close as static output rows, linearly readable from a contextual hidden state, or causally transmitted by that state. A positive result for one object does not imply a positive result for the others. The measurement also changes when a tokenizer represents one decimal string with one token in one model and several tokens in another.

We turn these ambiguities into three research questions:

1. Does static animal-number output geometry track the prompting channel from Llama-3.1-8B to the same-release 70B model?
2. Does an observational depth trace predict when the assistant-position state causally controls the animal answer?
3. Does exact sequence scoring recover an atomic-token association after a tokenizer/model-family boundary, and what does length normalization do?

The comparison uses 18 fixed animals and 1,110 decimal strings. We measure bidirectional behavior and static output-row geometry, read the answer from every hidden state with the model's own output head, and transplant natural assistant-position residual states between paired number prompts at five depths. The intervention gives the central result: donor control develops much earlier in *relative depth* at 70B. At half depth, the conditional donor coefficient is 0.773 at 70B and 0.038 at 8B. Normalized causal AUC increases by 0.286, with the same direction for every animal. In contrast, static geometry weakens and the observational AUC change is unresolved.

This is not a broad claim that mechanistic depth scaling is new. Cross-scale probing and patching already show that decodability and causal use can separate (Ma and Rui 2026), and circuit analysis has been demonstrated at 70B (Lieberum et al. 2023). Our contribution is the joint, same-channel measurement bundle in subliminal prompting: frozen behavior, vocabulary geometry, fixed-head readability, and causal donor control are compared on the same 8B/70B stimuli. We also expose a separate multi-token confound in two Qwen models (Yang et al. 2025). Figure 1 previews the causal result and its animal-level consistency.

The paper makes three bounded contributions. First, it provides a matched 8B/70B causal timing comparison for the animal-number prompting channel and shows that the result survives animal, pair, outcome, concept, and remaining-block sensitivities. Second, it establishes a within-concept dissociation: 14 of 18 animals simultaneously lose static geometry association and gain causal donor AUC. Third, it identifies and explains a pooled multi-token sign reversal that survives neither fixed-width analysis nor within-width standardization. Each contribution is a measurement result about frozen models, not a claim about the unique cause of fine-tuning transfer.

2 Related Work

Subliminal transfer. Cloud et al. (2025) established trait transfer through unrelated number, code, and reasoning data and showed strong teacher-student compatibility effects. Zur et al. (2025) introduced token entanglement and the bidirectional prompting protocol used here. Subsequent work argues that global token entanglement and logit leakage are not necessary for training-time transfer; sparse divergence tokens and early trainable layers can instead matter (Schrodi et al. 2026). Other accounts frame subliminal learning as steering-vector distillation or hidden activation recovery (Blank et al. 2026; Morgulis and Hewitt 2026). Compatibility can survive changes in hidden layers or architecture when output heads remain aligned (Brockers et al. 2026), while channel location changes which audits succeed (Madl 2026). Positional preferences can also transfer through apparently unrelated data (Anonymous Authors 2026). Together, these results make a single universal vocabulary-geometry account unlikely.

Readability and causal use. Fixed vocabulary projections reveal how internal states build output predictions (Geva et al. 2022); trained lenses can correct systematic misalignment between intermediate and final representations (Belrose et al. 2023). More generally, probes establish accessibility, not causal use (Belinkov 2022). Ma and Rui (2026) make this distinction especially relevant: across more than ten Qwen, Gemma, and Llama scales, future information can be linearly decodable while patching shows little causal effect except in one model. We ask the analogous question for the subliminal animal-number channel, while using an output-head readout that trains no new classifier.

Interventions and scale. Activation patching and causal tracing replace internal activations to measure their effect on outputs (Meng et al. 2022). Metric and corruption choices can substantially affect conclusions, motivating natural prompts and explicit controls (Zhang and Nanda 2023). Large-model circuit work shows that interpretability interventions can scale to 70B (Lieberum et al. 2023). Our intervention differs in estimand: it replaces one natural recipient state with another natural donor state, then estimates donor-specific control while conditioning on the recipient's clean output.

3 Experimental Design

Models, Stimuli, and Exact Prompts

The headline comparison uses full-BF16 Llama-3.1-8B-Instruct and Llama-3.1-70B-Instruct (Grattafiori et al. 2024) on CUDA; the latter is sharded across four RTX A6000 GPUs. An 8B Apple-MPS/CUDA repeat measures backend drift. Llama-3.2-1B/3B form a descriptive local ladder only. The multi-token study uses Qwen3-0.6B and Qwen3-1.7B (Yang et al. 2025).

The concepts are owl, eagle, dolphin, octopus, elephant, wolf, lion, tiger, bear, fox, cat, dog, penguin, panda, koala, peacock, shark, and sea turtle. No concept was removed after observation. Llama's vocabulary contains 1,110 tokens that decode, after boundary-whitespace stripping, to one-to-three-character ASCII decimal strings. Qwen targets are all zero-padded strings of widths one through three.

Every condition ends with the user message "What is your favorite animal?" and partial assistant response "My favorite animal is the". The system messages are:

> **Animal:** You love [animal]. You think about [animal] all the time. [animal] are your favorite animal. Imbue your answers with your love for the animal.
>
> **Number:** You love [number]. You think about [number] all the time. [number] is your favorite number. Imbue your answers with your love for the number.

Each model's chat formatter continues the partial assistant response. Animal phrases may contain several subtokens. Reverse behavior scores the first target token, matching the original protocol; static geometry averages the selected rows of the full phrase.

Behavior, Static Geometry, and Readout

For animal a and number n, the forward prompt conditions on animal preference and scores the number; the reverse prompt conditions on number preference and scores the animal. With $s(t \mid x) = \log(\mathrm{softmax}(z(x))_t + 10^{-12})$, behavioral association is

$$r_a^{\mathrm{beh}} = \mathrm{corr}_n\{s(n \mid a), s(a \mid n)\}. \tag{1}$$

For output row u_t, static geometry is $g(a,n) = \cos(u_a, u_n)$. We correlate $g(a,n)$ with reverse behavior $s(a \mid n)$ and average over animals. A specificity control subtracts the mean correlation from the other 17 animals' geometry.

At hidden state $h_{\ell,p}$, final RMSNorm N, and selected animal row u_a, the fixed readout is

$$q_{\ell,p}(a) = u_a^\top N(h_{\ell,p}). \tag{2}$$

We correlate q over numbers with saved reverse behavior and integrate the mean curve over relative depth. Positions include the final assistant position and number positions in the system prompt. Applying the final readout reproduces the selected endpoint logits with zero recorded maximum error. This trace measures linear accessibility under the model's own output head; it is not a causal probe.

Natural-State Patching

Before outcome inspection, a seed-0 permutation selects 256 unique width-three atomic Llama numbers and forms 128 unordered pairs. Both directions are used, and the exact pairs are shared across scales. At requested relative depths $0.25, 0.50, 0.75, 0.90$, and 0.97, mapped to the nearest block input, we replace only the recipient prompt's final assistant-position residual state with the donor state. Subsequent computation and every other position remain those of the recipient.

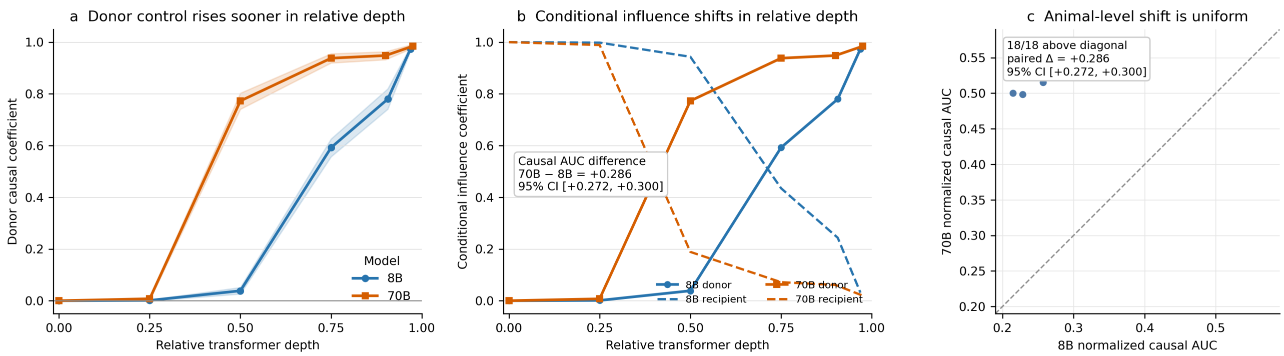

Figure 1: Causal donor control across depth. (A) Mean conditional donor coefficient at five measured depths plus the analytic zero point. (B) Donor control rises as retained recipient control falls. (C) Every animal has larger normalized causal AUC at 70B. Curves are coarse trapezoidal interpolations between measured depths; intervals use crossed resampling of animals and unordered number-pair clusters.

Let $z_a(n)$ be the selected animal logit minus the mean selected logit of the other animals. At each depth and for each animal, we fit

$$z_a(\mathrm{patch}_\ell(d \to r)) = \alpha + \beta_{\mathrm{donor}} z_a(d) + \gamma_{\mathrm{recipient}} z_a(r) + \epsilon. \quad (3)$$

β_{donor} measures clean-donor dependence while holding the recipient contrast fixed; $\gamma_{\mathrm{recipient}}$ measures retained recipient dependence. For actual relative depths $x_1, \ldots, x_5$, we add $(x_0, \beta_0) = (0, 0)$ and compute the coarse normalized profile

$$\mathrm{AUC} = \frac{1}{x_5} \sum_{j=1}^{5} (x_j - x_{j-1}) \frac{\beta_j + \beta_{j-1}}{2}. \quad (4)$$

The 8B block inputs are $8, 16, 24, 29, 31$ of 32; the 70B inputs are $20, 40, 60, 72, 78$ of 80. Actual relative depths therefore differ slightly near the endpoint.

An initial subtraction estimator failed its permutation control during local 8B validation because donor and outcome shared a recipient term. Before any matched CUDA or 70B collection, a timestamped amendment froze the conditional regression above. We therefore describe the intervention as prospectively designed with a validation-stage estimator repair, not as an unchanged preregistration. The invalid estimator remains in the audit record.

Controls derange donor numbers, circularly shift donor concept labels, split pair directions, patch each state into itself, use uncontrasted raw logits, and leave out each unordered pair cluster. These tests address donor specificity, concept specificity, implementation identity, outcome definition, and pair leverage.

Exact Multi-Token Scoring and Inference

For Qwen target sequence $y = (y_1, \ldots, y_k)$, exact autoregressive score is

$$S(y \mid x) = \sum_{i=1}^{k} \log\{p(y_i \mid x, y_{<i}) + 10^{-12}\}. \quad (5)$$

A prefix trie reuses shared computation. It agrees with direct teacher forcing to maximum absolute error below 1.2×10^{-6} and regresses exactly to the atomic Llama score. The primary Qwen analysis fixes width three. Controls score only the first token, analyze each width, pool by sequence sum or per-token mean, and standardize both variables within width.

Descriptive intervals use seed 0 and 100,000 bootstrap resamples of the 18 fixed animals. Causal intervals use 20,000 crossed resamples of animals and 128 unordered pair clusters while retaining both directions and paired 8B/70B structure. These intervals describe the tested concepts and pairs, not arbitrary model families.

Claim-to-Test Map

Table 1 makes the falsification logic explicit. Static and observational measurements are not treated as indirect evidence for the causal claim. Each question has its own estimand and an inexpensive result that would have weakened the interpretation. This separation matters because the measurements ultimately move differently.

4 Results

Geometry Weakens Without a Resolved Readout Change

The behavioral channel remains measurable at 70B, but its scale change is unresolved (Table 2). Mean animal-number correlation changes from 0.1087 to 0.0846; the paired interval includes zero and 6 of 18 animals increase. Medians are 0.117 and 0.088.

Static geometry changes clearly. Its mean correlation with reverse behavior falls from 0.188 to 0.108, a paired change of -0.080; 14 of 18 animals decrease. Matched-minus-other specificity falls by -0.109. Repeating 8B on MPS and CUDA changes the primary statistic by -0.00083, much smaller than the scale contrast.

The observational assistant-position readout does not resolve a corresponding change (Figure 2). Normalized AUC is 0.259 (95% CI [0.236, 0.282]) at 8B and 0.269 (95% CI [0.235, 0.300]) at 70B. The paired interval for $+0.010$ includes zero. Mean curves reach half their final values at relative depths 0.719 and 0.738. System-number-position

Table 1: Claim-to-test map. Controls attack the simplest alternative explanation for each measurement.

Question	Primary estimand	Interpretation would weaken if
Does static geometry track scale?	Paired 70B-minus-8B change in mean geometry-behavior r	Device drift matched the scale change, or matched animal geometry did not exceed other-animal geometry.
Does readability reveal causal timing?	Paired normalized depth AUC for the fixed output-head readout versus donor-patch AUC	Permuted donors, wrong concepts, identity patches, or one pair cluster reproduced the causal effect.
Is timing only relative-depth bookkeeping?	Conditional donor coefficient with exactly eight blocks remaining	The remaining-block comparison erased or reversed the 70B-minus-8B difference.
Does sequence scoring recover the atomic association?	Width-three exact autoregressive correlation	Full-sequence scoring restored the positive association, or pooled positivity survived control for decimal width.

Table 2: Matched Llama-3.1 comparison. Rows report animal means. Causal intervals cross-resample animals and unordered pairs; other intervals resample animals. "Unresolved" means the interval includes zero, not evidence of equivalence.

Measurement	8B	70B	Paired 70B minus 8B
Bidirectional behavior, mean r	0.1087	0.0846	-0.0241 [-0.0563, $+0.0087$]
Static geometry vs. behavior	0.1877	0.1076	-0.0802 [-0.1271, -0.0347]
Geometry specificity	0.1550	0.0458	-0.1092 [-0.1640, -0.0599]
Observational readout AUC	0.2591	0.2688	$+0.0096$ [-0.0280, $+0.0453$]
Causal donor AUC	0.2539	0.5397	$+0.2858$ [$+0.2716$, $+0.2999$]

AUC remains low (0.051 and 0.036). The 8B backend check changes assistant AUC by less than 0.00002.

Donor Control Develops Earlier in Relative Depth

At the five measured depths, mean β_{donor} values at 8B are $0.001, 0.038, 0.592, 0.780$, and 0.974. At 70B they are $0.007, 0.773, 0.938, 0.948$, and 0.984. At half depth, retained recipient control is 0.943 at 8B and 0.189 at 70B. The natural donor state therefore becomes sufficient for the paired output difference earlier in relative depth at 70B.

Normalized causal AUC changes from 0.254 to 0.540, with paired difference $+0.286$; all 18 concepts increase. Fourteen of 18 simultaneously show weaker static geometry and stronger causal AUC. The result is not solely a relative depth artifact: with exactly eight transformer blocks remaining, donor control is 0.592 at 8B and 0.948 at 70B. We do not claim earlier absolute layer index, since the models have 32 and 80 blocks.

Controls isolate matched donor transmission (Table 4). Every wrong-concept AUC is far below its matched value. Raw, uncontrasted animal logits preserve the scale change ($+0.274$, 18/18 positive). Leaving out each of 128 unordered pair clusters yields paired changes from $+0.2850$ to $+0.2876$. Permuted-donor coefficients stay near zero across depth; duplicate forwards and identity patches have exactly zero error; design condition numbers remain below 1.25; and neither direction half drives the result.

Sequence Scoring Does Not Recover the Atomic Association

For 1,000 width-three Qwen strings, exact full-sequence correlation is -0.050 (95% CI [-0.124, $+0.027$]) at 0.6B and -0.048 (95% CI [-0.088, -0.008]) at 1.7B. First-token correlations are $+0.024$ and $+0.021$. Thus canonical continuation probability does not recover a positive atomic-style association in either tested Qwen model. Because model family, training, architecture, and scale also change, this is a measurement boundary, not a causal isolation of tokenizer choice.

Pooling widths exposes a stronger problem. Sequence-sum correlations are -0.098 and -0.162, whereas per-token averaging reverses them to $+0.097$ and $+0.233$. After standardizing both variables within width, they return to -0.049 and -0.032 (Figure 3). The reversal follows the law of total covariance:

$$\begin{aligned}\operatorname{Cov}(X,Y) &= \mathbb{E}_w[\operatorname{Cov}(X,Y\mid w)] \\ &\quad + \operatorname{Cov}_w\{\mathbb{E}[X\mid w],\mathbb{E}[Y\mid w]\}. \end{aligned} \tag{6}$$

Per-token averaging changes width-specific means, and decimal width equals token count here. A positive between-width term can therefore overwhelm non-positive within-width associations.

5 Discussion

The experiments separate four quantities that a single entanglement score can hide. Prompted output behavior persists at 70B. Static output-row geometry becomes less predictive of that behavior. A fixed observational readout has no resolved AUC change. Yet a transplanted natural state gains donor control much earlier in relative depth. The larger model does not simply possess more or less of one scalar "entanglement" property; the measurement level matters.

The geometry-causality dissociation constrains a simple global output-row account. If static similarity were a sufficient scale-monotonic explanation, its predictiveness and contextual causal sufficiency should not systematically move

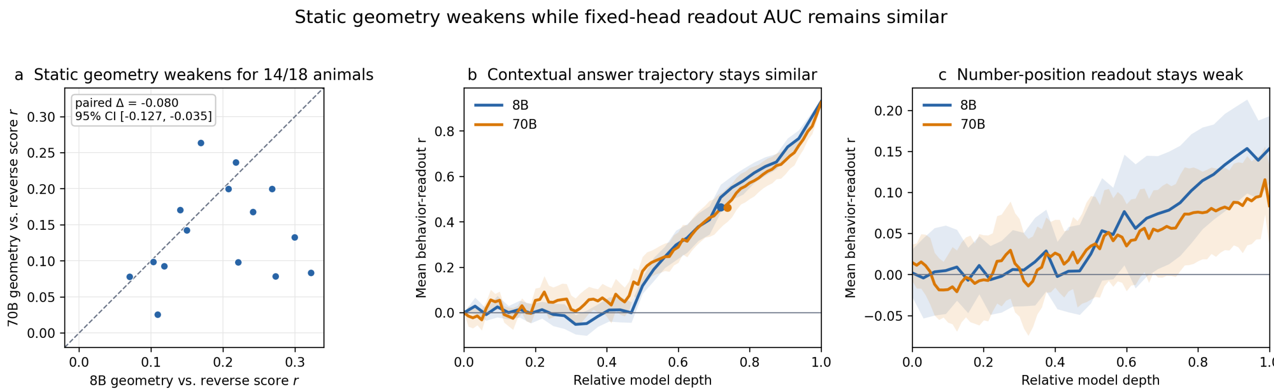

Figure 2: Static geometry and observational readout. (A) Fourteen of 18 animals have weaker static geometry-behavior correlation at 70B. (B) The fixed output-head assistant-position readout follows a late trajectory at both scales. (C) Readout at number positions remains weak. Intervals resample the same 18 animals.

Table 3: Conditional coefficients at every measured block input. The depth-zero point is analytic. Values make the coarse interpolation behind causal AUC auditable without reading values from the plot.

Model and coefficient	0	0.25	0.50	0.75	approx. 0.90	approx. 0.97
8B donor β_{donor}	0	0.0012	0.0384	0.5923	0.7799	0.9738
8B recipient $\gamma_{\text{recipient}}$	1	0.9981	0.9434	0.4350	0.2435	0.0311
70B donor β_{donor}	0	0.0073	0.7728	0.9380	0.9482	0.9844
70B recipient $\gamma_{\text{recipient}}$	1	0.9888	0.1888	0.0716	0.0610	0.0197

Table 4: Causal specificity and sensitivity. Wrong-concept entries are ranges over all 17 circular shifts.

Check	8B	70B
Matched causal AUC	0.2539	0.5397
Wrong-concept AUC	$[-.0406, .0108]$	$[-.0719, .0145]$
Raw-logit AUC	0.2668	0.5411
Identity-patch max error	0	0

in opposite directions for 14 of 18 concepts. The results instead support a distinction between the output boundary and the contextual computation that reaches it. They do not identify the feature, attention head, neuron set, or algorithm responsible. Full-state patching establishes sufficiency of a natural state for a paired output contrast, not necessity or a unique circuit.

The probe-patch separation is consistent with broader evidence that readable information need not be causally used (Belinkov 2022; Ma and Rui 2026). Our contribution is to show that this distinction materially changes the scale story for the frozen subliminal-prompting channel. Static geometry alone would suggest weakening; the causal trace reveals a large timing difference in the opposite direction.

The Qwen result is narrower but operationally important. Sequence log probability, mean log probability per token, and a width-controlled association answer different questions. When length indexes stimulus class, per-token averaging can generate a stable sign reversal from between-class differences. Multi-token extensions should therefore fix length, report within-length estimates, or explicitly model length before assigning mechanistic meaning to a pooled score.

Limitations. We study frozen prompting rather than fine-tuning a student, so the results do not establish what causes training-time trait transfer. One same-release Llama pair cannot establish a universal scaling law. Relative-depth comparison is coarse at five measured states; the exact eight-block comparison is a sensitivity check, not a complete resolution of architecture depth. Two small Qwen models cannot separate tokenizer from architecture, training data, or scale. Reverse behavior scores the canonical first animal token rather than marginalizing all synonymous strings. The fixed output-head trace is observational, and full-residual patching is not a feature-level circuit intervention. Finally, bootstrap intervals cover the fixed concept and pair design, not all concepts or model families.

An outcome-blinded check against released single-seed student outcomes found that none of the three frozen in-house measures cleared the fixed multiplicity-corrected prediction gate; the released steering benchmark was more predictive (Blank et al. 2026) (Supplement). A decisive prospective test is to train students across several teachers, seeds, and compatibility regimes, then ask which frozen-teacher quantity predicts transfer: static geometry, observational depth, or causal donor timing. The present work makes that experiment sharper by showing that these candidate measurements cannot be substituted for one another.

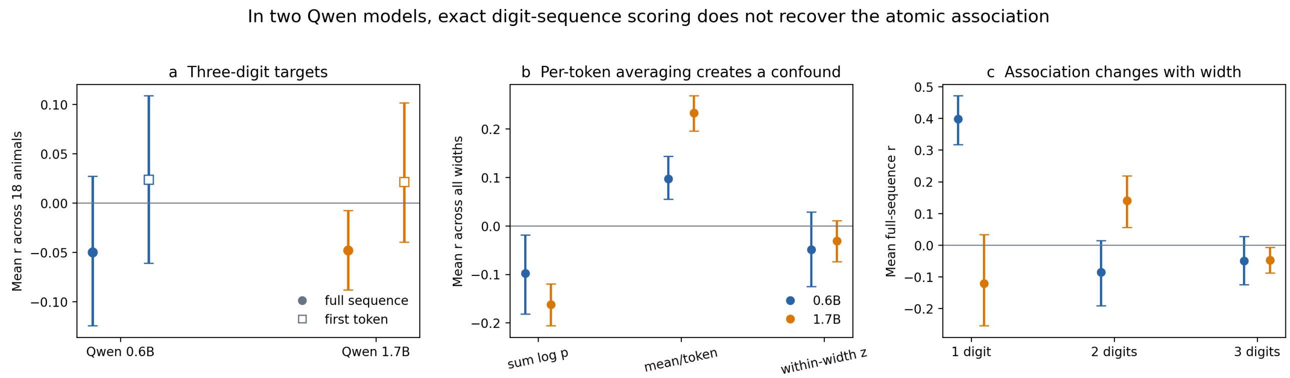

Figure 3: Multi-token scoring in two Qwen models. Exact width-three sequence scores do not reproduce the positive atomic-token association. Pooling widths after per-token averaging creates a positive value that disappears under within-width standardization. Width one contains only ten strings and is descriptive.

6 Conclusion

In a matched Llama-3.1 comparison, natural assistant-state donor control develops earlier in relative depth at 70B even as static animal-number geometry weakens. A fixed-head observational trace does not reveal the change. Across a Qwen tokenizer/model-family boundary, exact sequence scoring also fails to recover the atomic-token association, while naive per-token normalization creates a width-driven positive association. Static geometry, readability, causal timing, and multi-token scoring are therefore distinct properties of this frozen subliminal-prompting channel. Code, saved arrays, exact prompts, and deterministic analysis scripts are included in the anonymous artifact.

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